

CSE422: Artificial Intelligence (Lab Project)

Department of Computer Science & Engineering

BRAC University

Group No. 08

Project Title: Customer Category Classification using Machine Learning Techniques

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Table of Contents

Content	Page Number
Introduction	3
Dataset Description	3
Dataset Pre-processing	3 – 4
Dataset Splitting	4
Model Training & Testing	4
Model Comparison & Evaluation	5 - 6
Conclusion	6

Introduction

This project aims to build a machine learning system that automatically classifies customers into four distinct categories (A, B, C, and D) based on their demographic and behavioral data. The motivation is to help businesses identify target customer groups for marketing strategies and personalized services, thereby improving customer engagement and operational efficiency.

Dataset Description

The dataset consists of 8,068 data points with 10 input features and one target variable (**Segmentation**). This is a multi-class classification problem because the goal is to predict one of four discrete labels (A, B, C, or D) for each customer. There are two types of features in the given dataset “Customer Category Classification”. They are -

- Quantitative (Numerical): Age, Work_Experience, and Family_Size.
- Categorical: Gender, Ever_Married, Graduated, Profession, Spending_Score, and Var_1.

Categorical variables were encoded using LabelEncoder because machine learning models require numerical input to perform mathematical operations. A heatmap was generated to identify relationships between features. Features like Age and Spending_Score showed relevant patterns, though most features maintained independence, reducing redundancy. The target variable is fairly balanced. Segment D represents approximately 28.1% of the data, while A, B, and C make up 24.4%, 23.0%, and 24.4% respectively.

Dataset Pre-processing

During pre-processing, several faults were identified and resolved to ensure model stability:

- Null / Missing Values: Features like Work_Experience, Family_Size, and Profession contained missing entries.

Solution: We used SimpleImputer with the 'most_frequent' strategy to fill missing values of categorical columns and 'median' strategy for numerical columns, preserving the statistical distribution of the dataset.

- **Categorical Values:** Categorical data cannot be processed directly by algorithms like KNN or Neural Networks.

Solution: We applied Label Encoding to transform these into numerical representations.

- **Feature Scaling:** Numerical features had different scales (e.g., Age ranges from 18 to 89, while Family Size is 1 to 9).

Solution: We used StandardScaler to normalize the data, ensuring that features with larger ranges do not dominate the distance-based calculations in models like KNN. Initially, we also applied MinMaxScaling but the accuracy rate were less than the current rate for KNN and Decision tree but we had a boost on the accuracy rate of the Neural Network (around 53.2%)

Dataset Splitting

The dataset was split using a Random split strategy:

- Train set: 80% (6,454 samples) used for training the models.
- Test set: 20% (1,614 samples) used for final performance evaluation.

Model Training & Testing

We implemented and compared four different approaches:

1. **KNN (Supervised):** A distance-based classifier used to group customers based on similarity.
2. **Decision Tree (Supervised):** A tree-based model that splits the data based on feature importance to reach a classification decision.
3. **Neural Network (Supervised):** Used an MLPClassifier (Multi-Layer Perceptron) with hidden layers to capture complex, non-linear relationships in the customer data.
4. **K-Means (Unsupervised):** To satisfy the unsupervised learning requirement, we applied K-Means to identify clusters. This helped visualize how customers group naturally without using the target labels.

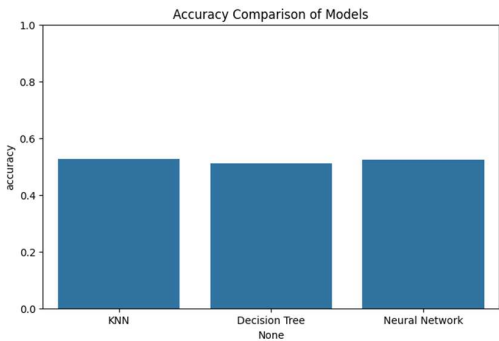
Model Selection / Comparison Analysis

Performance Metrics: All models were evaluated based on Accuracy, Precision, Recall, and ROC/AUC scores. The following image shows the result achieved by each of the models.

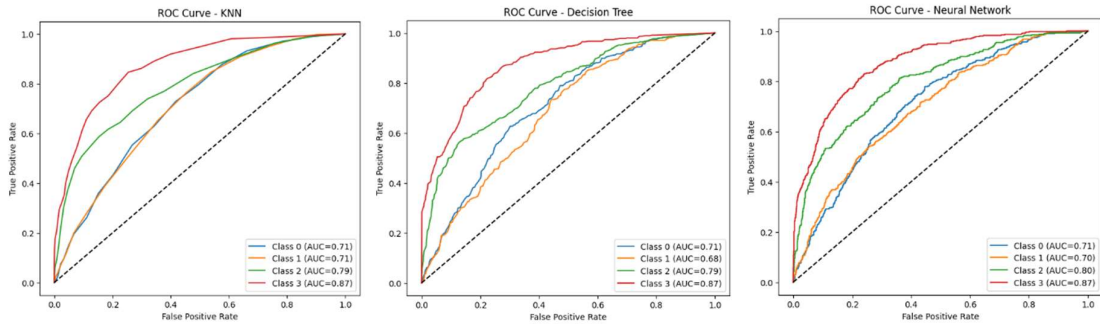
```
Training Model: KNN
KNN Accuracy: 0.5266 (52.66%)
KNN Precision: 0.5155 (51.55%)
KNN Recall: 0.5266 (52.66%)

Training Model: Decision Tree
Decision Tree Accuracy: 0.5118 (51.18%)
Decision Tree Precision: 0.5058 (50.58%)
Decision Tree Recall: 0.5118 (51.18%)

Training Model: Neural Network
Neural Network Accuracy: 0.5229 (52.29%)
Neural Network Precision: 0.5118 (51.18%)
Neural Network Recall: 0.5229 (52.29%)
```



The following images are the ROC Curves of the models, the AUC scores are also written –

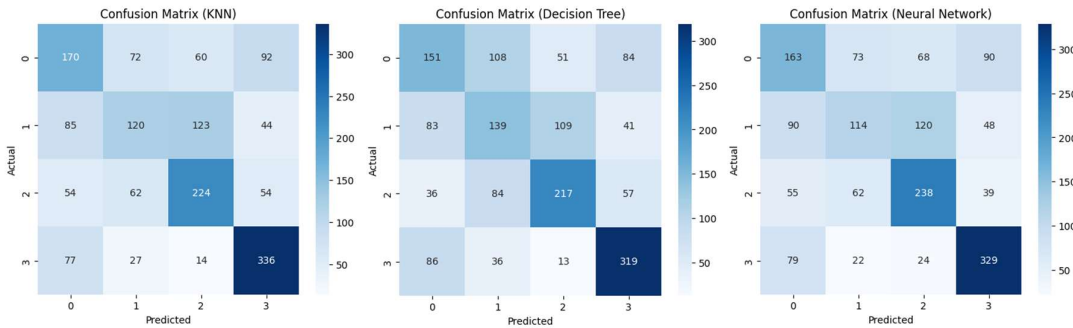


KNN Mean AUC Score: 0.7700

Decision Tree Mean AUC Score: 0.7628

Neural Network Mean AUC Score: 0.7711

The following image is the Confusion Matrix –



Results Summary -

KNN emerged as the Best Performing Model, effectively utilizing the distance-based relationships between customer features. The Neural Network achieved strong performance, showing its ability to learn complex patterns, although it required more computational time. Decision Tree performed well but was slightly more sensitive to feature overlap. Bar charts were used to compare accuracy across models, and ROC curves were plotted for each class to visualize the trade-off between sensitivity and specificity.

Conclusion

Findings: The project demonstrated that demographic data is a strong predictor of customer segments. KNN performed best, suggesting that customers in the same segment tend to have very similar feature values.

Challenges: Dealing with missing values in categorical columns like Profession was challenging, as assigning the most frequent value might introduce slight bias. Too many categorical features than numerical features was also a challenging part for model training.

Proper feature scaling was the most essential step. Without it, the KNN model's performance was significantly lower. Future work could involve hyperparameter tuning to further improve the Neural Network's accuracy and improving dataset quality by introducing new data to improve accuracy of all the models.